

Safety-Aware Elevation of Privilege

ICS Threat-Modeling Game — Interface Showcase

A safety-aware modification of the Elevation of Privilege card game for Smart Grid ICS threat modeling, anchored in a Purdue-model architecture with an explicit Safety/HSE impact dimension on every threat.

The following screenshots were captured from the **working local demonstrator**. They show the developed interfaces (modifications M1–M5). Work in progress — for internal sharing.

Generated 15 June 2026

Connect screen

Local-first backend selection. The realtime server runs on this machine/LAN — no internet required. Hosted Firebase play is marked **Coming soon**.

🔌 Connect to server

Choose the realtime backend

Pick where authentication, rooms, player joins, submissions, timer updates, and debrief state will live for this run.



Connect local server

Use the local realtime database and local auth service on this machine.

Local server URL

http://127.0.0.1:8788

Connect local server

Start it with `npm run server:local`. Default database: `.local-realtime-db.json`.



COMING SOON

Firebase online server

Hosted multi-device play over the internet is in progress. For now, use the local server — it runs the full workshop on one machine or LAN.

Coming soon

Home — two clear paths

Play (join a game as a participant) versus **Facilitate** (create and run a room). One account joins one game *or* facilitates one room; the app shows only what each role needs.

CONNECTED SERVER
Local server - C:\Users\persian\AppData\Local\Temp\oop-shots-db...

Change server

NOT CONNECTED
Not connected
Sign in from Home to manage rooms.

NOT CONNECTED
Login

Safety-aware EoP

What do you want to do?

Choose one path. The app will show only what is needed for that role.

Local mode - Firebase paused

WORKING LOCALLY
Firebase Auth is paused. Local mode uses the realtime server on this machine, so other browsers can join when they use the same local server URL.

Try Firebase again

PLAY

Join a game

Take part as a participant. You need the room code and the PIN your facilitator shares with you (by email or on the projected screen).

Join with code + PIN

Browse active rooms

FACILITATE

Manage a room

Create and run your own game as facilitator. One account manages one room at a time.

Login to facilitate

The game in play — guided board (M1, M2, M5)

Always-visible board for someone who never read the paper: the explicit attacker goal (M2), the Smart Grid Purdue target (M1), one-card-per-STRIDE rotation (M5), the six-step turn loop, and the structured threat entry.

CONNECTED SERVER
Local server - C:\Users\persian\AppData\Local\Temp\ueop-shots-db.json

Change server

CONNECTED ACCOUNT
Workshop Facilitator
e2e-1781556258477-2xlm@example.local

FACILITATOR

Disconnect

MANAGE

Safety EoP workshop room

Code 5HFQH4 - active

Home

Safety-aware Elevation of Privilege for ICS

Safety-Aware EoP for Smart Grid ICS

A fast, social threat-modeling game for Smart Grid Purdue architectures where every cyber threat must also be judged for Safety/HSE impact.

ATTACKER GOAL

Create the most impactful threat leading to a safety issue for people and society by making electricity unavailable for the DSO that controls the smart grid.

59:59
Workshop timer

0
Accepted threats
Structured outputs

0
High/Critical cyber-safety
Prioritise in debrief

0/6
STRIDE coverage
Use rotation to fill gaps

Home

Manage

Play

Screen

Deck

Architecture

Register

Debrief

Setup

Guide

YOUR MISSION — YOU ARE THE ATTACKER

Create the most impactful threat leading to a safety issue for people and society by making electricity unavailable for the DSO that controls the smart grid.

Active player: Workshop Facilitator

Pause game

How to play

ROUND 1 - CARD 1 OF 6

One card per STRIDE category

STRIDE

SpooFing

Can an attacker pretend to be a legitimate user, device, service, or control signal?
Think credentials, device identity, VPN access, HMI operator sessions, RTU/IED identity.

EACH TURN

1 Read the STRIDE card

2 Pick a component on the map

3 Describe the attack

4 Group discusses

5 Rate cyber + Safety/HSE

6 Record / End turn

Current turn

Workshop Facilitator

One card per STRIDE category

Rotation

Active decks: Custom 1 - Smart Grid safety

STRIDE

SpooFing

Fake operator session

An attacker can reuse or guess operator credentials and appear as a legitimate SCADA user.

Example: Apply to SCADA HMI, VPN Access Server, or Office Workstation.

Facilitator prompt: Can an attacker pretend to be a legitimate user, device, service, or control signal?

New card

End turn

Smart Grid Purdue model

Pick a target component or connection before accepting a threat.

Research paper architecture diagram

Supports PNG, JPG, SVG, or PDF. The preview is stored in this browser.

Upload diagram

No diagram uploaded yet. Export or screenshot the architecture figure from your paper, then upload it here.

Office network

Layer 0

Office Workstation

Office Router

Business Server

Shared Folder

Identity Provider

Office to DMZ trust boundary

Office <-> DMZ

Office Workstation <-> Jump Host

Identity Provider <-> Jump Host

DMZ

Layer 1

Data Exchange Server

DMZ Firewall

Jump Host

Patch Repository

Historian Mirror

DMZ to SCADA trust boundary

DMZ <-> SCADA

Data Exchange Server <-> SCADA Server

Historian Mirror <-> SCADA Server

Jump Host <-> SCADA HMI

SCADA network

Layer 2

SCADA Server

SCADA HMI

MTU

VPN Access Server

SCADA Firewall

SCADA to OT control path

SCADA Server <-> MTU

VPN <-> SCADA

MTU <-> RTU

Field Technician <-> HMI

OT / field network

Layer 3

RTU

IED

PLC

Process Network

Field Switch

MV/LV Bus

Primary Asset

Protection Relay

Field-level connections

RTU <-> IED

IED <-> Protection Relay

PLC <-> Process Network

Field Switch <-> Primary Asset

+ Record accepted threat

A threat is complete only when likelihood, cyber impact, target, and Safety/HSE impact are discussed.

Threat name

Example: False process view on SCADA HMI

Description

In this context, an attacker could...

STRIDE category

SpooFing

Device / connection link

SCADA Server - SCADA network

Likelihood

Medium

Cyber impact

Medium

Safety/HSE impact +

Choose Safety/HSE impact +

Required: what physically happens if this succeeds?

Calculated indication for debrief

Cyber risk: Medium (4)

Safety risk: set the tag

Cyber safety: —

The score is a workshop indication: risk = likelihood x impact, then cyber-safety = cyber risk x safety risk.

Mitigation idea

Controls, monitoring, procedure, training...

Player / group

Workshop Facilitator

Group accepted the threat

Add to threat register

Set the Safety/HSE impact to continue.

Threat register

Structured output for debrief, prioritisation, and export.

CSV

JSON

Search threats...

All

All

No threats match the current filters.

Projector / join view

The facilitator projects this: the room **code**, the **PIN**, and a join **QR** that carries the PIN for automatic join. Players scan to join, or type the code + PIN.

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Change server

CONNECTED ACCOUNT
Workshop Facilitator
e2e-1781556258477-2x1m@example.local

FACILITATOR

Disconnect

MANAGING

Safety EoP workshop room

Code SHFQH4 · active

Home

Safety-aware Elevation of Privilege for ICS

Safety-Aware EoP for Smart Grid ICS

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ATTACKER GOAL

Create the most impactful threat leading to a safety issue for people and society by making electricity unavailable for the DSO that controls the smart grid.

59:57
Workshop timer

Pause

0

Accepted threats

Structured outputs

0

High/Critical cyber-safety

Prioritise in debrief

0/6

STRIDE coverage

Use rotation to fill gaps

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ROOM SCREEN

Room code

Timer

Participants

Accepted threats

CURRENT CARD

Fake operator session

An attacker can reuse or guess operator credentials and appear as a legitimate SCADA user.

Category: Spoofing

Join this room



<http://127.0.0.1:5174/?room=SHFQH4&role=player&pin=7421>

ROOM CODE

SHFQH4

PIN

7421

Scan the QR to join automatically, or open the app, choose Join room, and enter the code + PIN.

No PIN? You can still join — the facilitator will approve you manually.

0

High/Critical cyber-safety

Discuss first

0

Waiting approval

Approval enabled

Structured threat register (M3, M4)

Every threat carries the **unskippable Safety/HSE tag** (M3) inside the structured template (M4) — name, description, STRIDE category, cyber severity, device/connection link — with computed cyber / safety / cyber-safety risk.

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59:44
Workshop timer

Pause

5
Accepted threats
Structured outputs

3
High/Critical cyber-safety
Prioritise in debrief

4/6
STRIDE coverage
Use rotation to fill gaps

Threat register

Structured output for debrief, prioritisation, and export.

CSV

JSON

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Spoofing

Likelihood: Medium

Medium

High

Cyber risk: Medium (4)

Safety risk: High (6)

Cyber-safety: High (6)

SCADA Server

Accepted

Delete

Alter PLC safety setpoints

Change protection limits so unsafe operation is accepted as normal.

By Workshop Facilitator · 6/15/2026, 10:44:28 PM

Denial of Service

Likelihood: Medium

Medium

Medium

Cyber risk: Medium (4)

Safety risk: Medium (4)

Cyber-safety: Medium (4)

SCADA Server

Accepted

Delete

Flood the data-exchange gateway

Saturate the DMZ gateway so operators lose visibility of the field.

By Workshop Facilitator · 6/15/2026, 10:44:27 PM

Elevation of Privilege

Likelihood: Medium

Medium

High

Cyber risk: Medium (4)

Safety risk: High (6)

Cyber-safety: High (6)

SCADA Server

Accepted

Delete

Workstation to SCADA-server escalation

Pivot from an office workstation onto the SCADA server (IT-to-OT path).

By Workshop Facilitator · 6/15/2026, 10:44:27 PM

Spoofing

Likelihood: Medium

Medium

Medium

Cyber risk: Medium (4)

Safety risk: Medium (4)

Cyber-safety: Medium (4)

SCADA Server

Accepted

Delete

Fake operator session on the SCADA HMI

Reused operator credentials let an attacker issue commands as a trusted user.

By Workshop Facilitator · 6/15/2026, 10:44:26 PM

Tampering

Likelihood: Medium

Medium

High

Cyber risk: Medium (4)

Safety risk: High (6)

Cyber-safety: High (6)

SCADA Server

Accepted

Delete

Replay on the SCADA-MTU-RTU command path

Control traffic without nonces can be replayed to mis-operate a breaker.

By Workshop Facilitator · 6/15/2026, 10:44:25 PM

Debrief — dashboards & risk matrices

Top cyber-safety threats, STRIDE coverage, Safety/HSE distribution, and the cyber / safety / cyber-safety risk matrices for the cross-group debrief.

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FACILITATOR

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MANAGING

Safety EoP workshop room
Code SHFQH4 - finished

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59:43
Synced room timer

Start

5
Accepted threats
Structured outputs

3
High/Critical cyber-safety
Prioritise in debrief

4/6
STRIDE coverage
Use rotation to fill gaps

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Top cyber-safety threats

#1 Alter PLC safety setpoints

Cyber-safety: High (6)

Cyber risk: Medium (4)

Safety risk: High (6)

Likelihood: Medium

SCADA Server - Spoofing

#2 Workstation to SCADA-server escalation

Cyber-safety: High (6)

Cyber risk: Medium (4)

Safety risk: High (6)

Likelihood: Medium

SCADA Server - Elevation of Privilege

#3 Replay on the SCADA-MTU-RTU command path

Cyber-safety: High (6)

Cyber risk: Medium (4)

Safety risk: High (6)

Likelihood: Medium

SCADA Server - Tampering

#4 Flood the data-exchange gateway

Cyber-safety: Medium (4)

Cyber risk: Medium (4)

Safety risk: Medium (4)

Likelihood: Medium

SCADA Server - Denial of Service

#5 Fake operator session on the SCADA HMI

Cyber-safety: Medium (4)

Cyber risk: Medium (4)

Safety risk: Medium (4)

Likelihood: Medium

SCADA Server - Spoofing

Debrief dashboard

STRIDE coverage

S Spoofing

T Tampering

R Repudiation

I Information

D Denial

E Elevation

2

1

0

0

1

1

Safety/HSE impact distribution

High

Medium

Low

3

2

0

Cyber-safety score distribution

Critical

High

Medium

Low

0

3

2

0

Facilitator questions

Which Purdue layers attracted the most High or Critical cyber-safety scores?

Which STRIDE categories were neglected?

Which threats have low cyber risk but high safety risk?

Which mitigations are technical, procedural, or training-based?

Cyber risk matrix

Likelihood x cyber impact. This captures the cyber-security risk before safety consequence is considered.

Likelihood / Cyber impact	Low	Medium
High	Medium Score 3	High Score 6
Medium	Low Score 2	Medium Score 4 5 threats
Low	Low Score 1	Low Score 2

Safety risk matrix

Likelihood x Safety/HSE impact. This captures the physical consequence path after the cyber scenario is described.

Likelihood / Safety impact	Low	Medium
High	Medium Score 3	High Score 6
Medium	Low Score 2	Medium Score 4 2 threats
Low	Low Score 1	Low Score 2

Cyber-safety score

Cyber risk x safety risk. Use this only as a debrief indication for prioritising discussion.

Safety risk / Cyber risk	Low	Medium
High	Medium Score 3	High Score 6 3 threats
Medium	Low Score 2	Medium Score 4 2 threats
Low	Low Score 1	Low Score 2

Smart Grid Purdue target (M1)

The shared, editable Purdue-model architecture every threat is anchored to — Office / DMZ / SCADA / OT field zones.

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Office network

Layer 0

Office Workstation

Office Router

Business Server

Shared Folder

Identity Provider

Office to DMZ trust boundary

Office <-> DMZ

Office Workstation <-> Jump Host

Identity Provider <-> Jump Host

DMZ

Layer 1

Data Exchange Server

DMZ Firewall

Jump Host

Patch Repository

Historian Mirror

DMZ to SCADA trust boundary

DMZ <-> SCADA

Data Exchange Server <-> SCADA Server

Historian Mirror <-> SCADA Server

Jump Host <-> SCADA HMI

SCADA network

Layer 2

SCADA Server

SCADA HMI

MTU

VPN Access Server

SCADA Firewall

SCADA to OT control path

SCADA Server <-> MTU

VPN <-> SCADA

MTU <-> RTU

Field Technician <-> HMI

OT / field network

Layer 3

RTU

IED

PLC

Process Network

Field Switch

MV/LV Bus

Primary Asset

Protection Relay

Field-level connections

RTU <-> IED

IED <-> Protection Relay

PLC <-> Process Network

Field Switch <-> Primary Asset